

CIS 671 Final Project Design

Project Title: Global Electric Vehicle (EV) Adoption & Market Trends Visualization

Team Members:

- Karthika Ganjilenka
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Dataset Description:

Source: Kaggle

Dataset: Global EV Sales 2010-2024 Dataset

Visual Components Overview

We will implement 6 visual components (3 per team member) using D3.js for visualization and Python (Pandas) for data preprocessing. The following are the proposed components:

For Karthika Ganjilenka:

Line Chart: Global EV Sales Growth Over Time

Bar Chart: Regional EV Sales Trends Comparison

Stacked Bar Chart: Powertrain Type Analysis (BEV vs. PHEV)

For Devendhar Sangraj:

Treemap: Regional Cumulative EV Sales Comparison

Choropleth Map: Country-Wise EV Penetration Levels

Pie Chart: Powertrain Market Share for Specific Year & Region

(Optional visualization if time permits)

Scatter Plot: Correlation Between EV Sales & CO2 Emissions

Type of Charts, Interactions, and Rationale

Line Chart – Global EV Sales Growth Over Time

Purpose: Track global EV sales growth from 2010 to 2024.

Interaction: Hover tooltips to show exact values, zoom/pan for detailed exploration.

Rationale: Effective for understanding long-term sales growth patterns.

Bar Chart – Regional EV Sales Trends Comparison

Purpose: Compare regional sales growth trends.

Interaction: Filter by year or powertrain type.

Rationale: Facilitates side-by-side comparisons across regions.

Stacked Bar Chart – Powertrain Type Analysis (BEV vs. PHEV)

Purpose: Visualize the proportion of BEV and PHEV sales by region.

Interaction: Filter by year and region.

Rationale: Highlights regional preferences for electric vehicle types.

Treemap – Regional Cumulative EV Sales Comparison

Purpose: Visualize cumulative EV sales across regions using nested rectangles, sized by sales volume and divided by powertrain type (BEV or PHEV).

Interaction: Interactive filtering by region and powertrain type with hover details and zoom functionality.

Rationale: Treemaps provide a clear, space-efficient comparison of regional EV sales, making it easy to identify top-performing regions.

Choropleth Map – Country-Wise EV Penetration Levels

Purpose: Visualize the geographical distribution of EV penetration rates.

Interaction: Clickable country-level details and region-based filtering.

Rationale: Geospatial patterns are best represented using maps.

Pie Chart – Powertrain Market Share for Specific Year & Region

Purpose: Display the percentage split between BEV and PHEV.

Interaction: Filter by year and region.

Rationale: Effective for showing part-to-whole comparisons.

Visual Encoding Choices and Justification

Line Chart: Time on the x-axis, sales on the y-axis, color-coded lines for each region.

Bar Chart: Regions on the x-axis, sales on the y-axis, color-coded bars for powertrain types.

Stacked Bar Chart: Region on the x-axis, cumulative sales on the y-axis, segments representing BEV and PHEV.

Treemap: Nested rectangles sized by cumulative EV sales, with color representing region and subdivisions indicating BEV and PHEV sales. Larger rectangles signify higher sales volumes.

Choropleth Map: Color gradient representing EV penetration percentage per country.

Pie Chart: Slices representing BEV and PHEV share, size proportional to market share.

These encodings are justified by the nature of the data, providing clarity, interpretability, and actionable insights for users.

Sketches and Mockups

Below are basic descriptions of how the visualizations will appear.

Line Chart: A multi-line chart with separate-colored lines representing different regions, showing growth trends over time.

Bar Chart: Vertically aligned bars grouped by region, using color to differentiate powertrain types.

Stacked Bar Chart: Stacked horizontal bars with cumulative EV sales, categorized by BEV and PHEV.

Treemap: Nested rectangles representing regions sized by cumulative EV sales, subdivided by powertrain type (BEV and PHEV). Color gradients indicate relative sales volume, with tooltips displaying detailed information.

Choropleth Map: World map with shaded regions indicating penetration levels using a gradient scale.

Pie Chart: Circular chart segmented to represent market share, with tooltips displaying detailed information.

Conclusion

This design ensures clear data-driven insights using appropriate visualization techniques. With effective interactivity and clean visual encoding, the proposed dashboard will provide a comprehensive understanding of global EV adoption trends for policymakers, analysts, and researchers.

Next Steps:

Begin preprocessing the dataset using Python.

Implement visualizations using D3.js.

Conduct iterative feedback and refinement cycles.